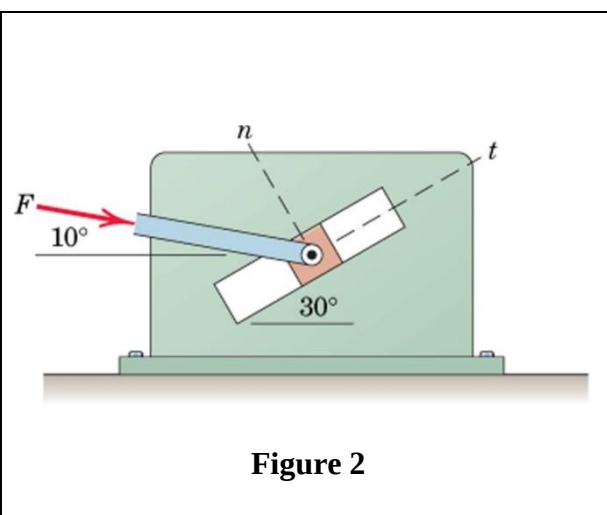
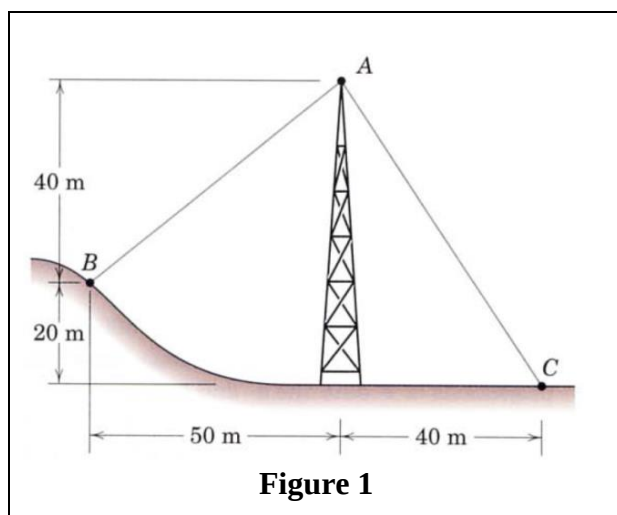
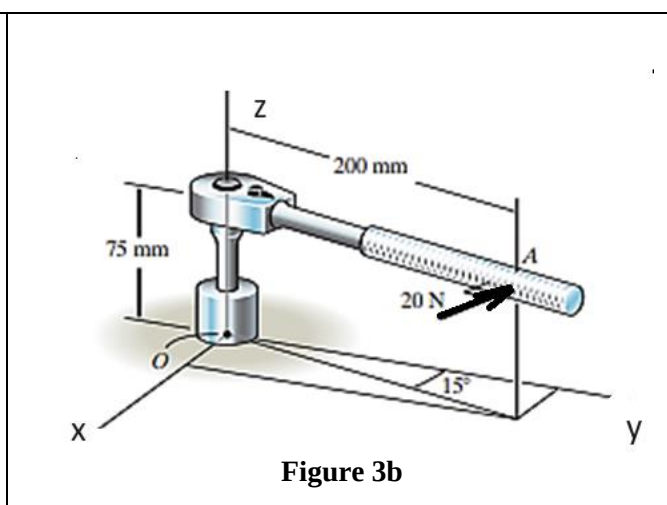
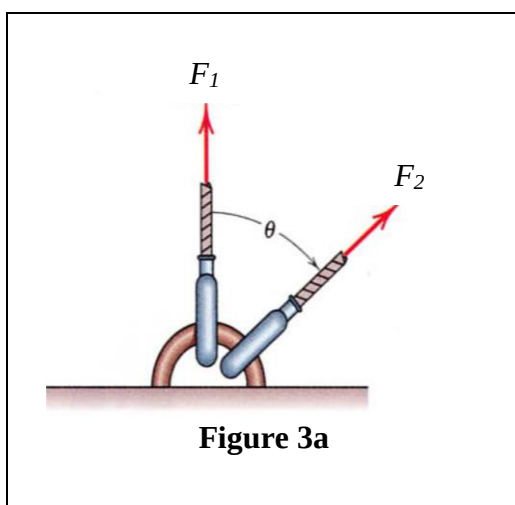


Tutorial-1
ME-101, (Engineering Mechanics, Semester-II)
Date: 18/01/2024 and Time: 7.55 to 8.50 AM

1. The guy cables AB and AC are attached to the top of the transmission tower as shown in **Figure 1**. The tension in the cable AC is 8 kN. Determine the required tension in cable AB such that the net effect of the two cable tensions is a downward force at point A . Determine the magnitude R of the downward force . **[10 marks]**



2. The t component of the force F is known to be 75 kN and is shown in **Figure 2**. Determine the n - component and the magnitude of F . **[10 marks]**
3. a) Two forces F_1 and F_2 act on the hook (**Figure 3a**). If their lines of action are at an angle θ apart and the magnitude of each force is $F_1 = F_2 = F$, determine the magnitude of the resultant force F_R and the angle between F_R and F_1 . **[5 marks]**
- b) A 20 N horizontal force is applied perpendicular to the handle of the socket wrench (refer **Figure 3b**). Determine the magnitude and the coordinate direction angles of the moment created by this force about point O . **[5 marks]**



4. While sliding a desk, three students exert the forces as shown in the overhead view in **Figure 4**. Determine the equivalent force—couple system at point **A**. Also determine the equation of the line of action of the resultant force. [5+5= 10 marks]

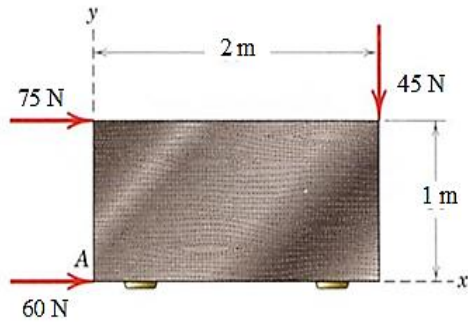


Figure 4

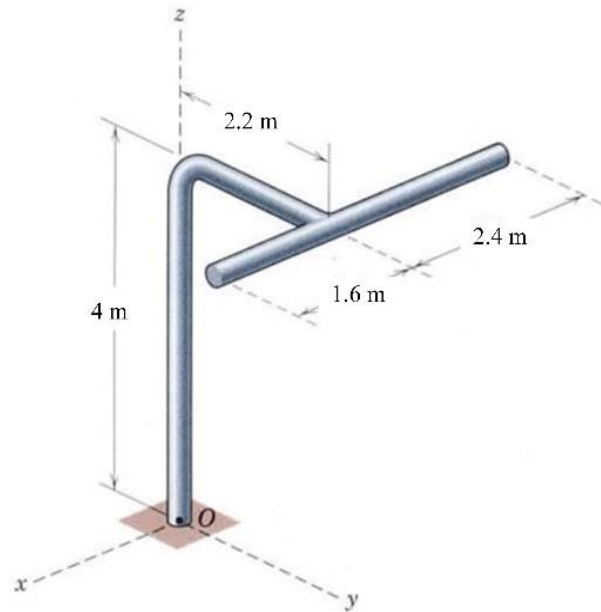


Figure 5

5. The structure as shown in **Figure 5** is constructed of circular rod which has a mass of 7 kg per meter of length. Determine the moment M_O about O caused by the weight of the structure. Also find the magnitude of M_O . [5+5= 10 marks]
6. A uniform wheel of 800 mm diameter of weight 8 kN rests against a rigid rectangular block of 200 mm height as shown in **Figure 6**.

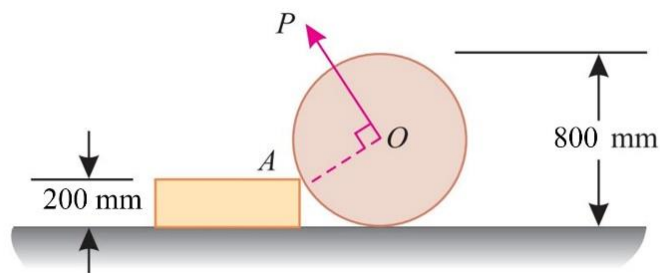


Figure 6

Find the least pull, through the centre of the wheel, required just to turn the wheel over the corner **A** of the block. Also find the reaction on the block at **A**. Consider all the surfaces to be smooth i.e. frictionless. [5+5= 10 marks]